Regime-Adaptive Multimodal Transformer for Cross-Market Equity Forecasting

HMM Regime Detection, Feature Engineering, and XGBoost Baseline

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Abstract—This report presents Phase-1 deliverables for cross-market equity forecasting using one U.S. stock (JPM) and four Indian stocks (RELIANCE, TCS, HDFCBANK, EPIGRAL). The current phase focuses on exploratory diagnostics, hidden Markov model (HMM) regime detection, a 36-column engineered feature table, and a walk-forward XGBoost baseline. The objective is to establish technically reliable foundations before introducing heavier deep architectures. We enforce temporal causality using T-1 time-zone alignment for cross-market linkage features to avoid look-ahead bias. Results show moderate directional performance with ticker-dependent risk-adjusted outcomes, motivating multimodal extension in later phases.

Index Terms—equity forecasting, hidden Markov model, regime detection, feature engineering, XGBoost, cross-market learning

I. Introduction

Cross-market forecasting between U.S. and Indian equities is sensitive to regime shifts, market microstructure differences, and asynchronous trading sessions. In this project, the modeling universe is JPM (NYSE) and RELIANCE/TCS/HDFCBANK/EPIGRAL (NSE). The immediate Phase-1 requirement is not architectural complexity, but methodological correctness: robust data preprocessing, regime labeling, leakage-safe alignment, and a reproducible baseline.

Our exploratory analysis indicates heavy-tailed returns, volatility clustering, and time-varying stock-index coupling. These observations motivate regime-aware feature construction. Accordingly, the Phase-1 pipeline prioritizes: (i) consistent return/volatility transformations, (ii) explicit HMM state inference, (iii) cross-asset context via rolling market correlation, and (iv) chronological walk-forward validation for baseline benchmarking.

Gap Argument (future direction). Existing SOTA directions are strong but fragmented: hybrid Transformer-GNN models improve dependence forecasting [6], and LLM-driven volatility-adaptive systems improve multimodal alignment [7]. However, a unified US–India transfer framework that combines explicit HMM regimes with multimodal transformer gating is a later-phase target. Phase-1 therefore establishes the verified statistical and feature-engineering base needed for that step.

II. Literature Review

A. Classical Time-Series Limits Under Regime Shifts

ARIMA/GARCH families are foundational but assume relatively stable generating dynamics over calibration windows [1]–[3]. In equity data, abrupt stress periods can shift volatility and dependence faster than static parameterizations can adapt.

B. Cross-Market Dependence and Relational Modeling

Fanshawe et al. show that hybrid temporal-relational models improve forward dependence estimation and downstream portfolio structure, including residual modeling in Fisher-z space [6]. This supports explicit treatment of cross-asset connectivity rather than purely univariate forecasting.

C. Multimodality and Volatility Adaptation

Ying et al. highlight that volatility-adaptive systems benefit from stronger cross-modal alignment between textual and price signals [7]. While Phase-1 here is price-feature focused, this literature informs future extension beyond numerical factors.

D. Critical Synthesis and Remaining Gap

While each of these methods addresses a specific limitation, no existing framework simultaneously combines explicit HMM-derived regime supervision, multimodal feature fusion, and cross-market transfer learning. Fanshawe et al. model dependence structure effectively, including residual Fisher-z stabilization, but do not provide explicit regime-gated transfer logic across asynchronous markets [6]. Ying et al. strengthen multimodal alignment under volatility stress, but do not target cross-market transfer with externally supervised regime states [7]. This leaves a clear methodological gap: a unified system that is regime-explicit, multimodal, and transfer-aware. The proposed RAMT direction is intended to fill this exact intersection in later phases.

III. Methodology (Phase-1)

A. Data Universe and T-1 Time-Zone Alignment

The universe is:

- JPM (U.S., NYSE)
- RELIANCE.NS, TCS.NS, HDFCBANK.NS, EPIGRAL.NS (India, NSE)

To avoid look-ahead bias in cross-market signals, U.S.-linked information used for India-facing prediction is lagged by one decision day (T-1 protocol).

B. HMM Regime Detection

For each ticker, we fit a 3-state Gaussian HMM on $[\text{Log_Return}, \text{Realized_Vol}_{20}]$, then map latent states to semantic regimes $\{\text{bull}, \text{bear}, \text{high-vol}\}$.

State transition matrix:

$$A = \{a_{ij}\}, \quad a_{ij} = P(s_t = j \mid s_{t-1} = i), \quad \sum_{j=1}^{K} a_{ij} = 1. \tag{1}$$

Gaussian emission likelihood for state j:

$$b_j(\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_j, \Sigma_j), \tag{2}$$

where $\mathbf{x}_t = [r_t, \sigma_t^{(20)}]^\top$, r_t is log-return, and $\sigma_t^{(20)}$ is 20-day realized volatility.

C. 36-Column Feature Engineering Pipeline

In features/feature_engineering.py, the processed table is built from seven feature groups:

1) Lagged returns: Return_Lag_{1,2,3,5,10,20}
2) Volatility: Realized_Vol_{5,20,60}, Garman_Klass_Vol, Vol_Ratio
3) Technicals: RSI, MACD stack, Bollinger features
4) Momentum: Momentum_{5,20,60}, ROC_10
5) Volume: Volume_MA_Ratio, Volume_Log
6) Regime features: HMM_Regime, HMM_Regime_Label
7) Cross-asset context: Rolling_Corr_Index

The cross-asset correlation uses ĜSPC for JPM and ℕSEI for Indian names.

D. XGBoost Baseline Protocol

Phase-1 baseline uses models/baseline_xgboost.py with:

- chronological target: next-day log return,
- expanding-window walk-forward evaluation,
- early stopping with validation split inside each fold,
- regime-stratified experts (when sufficient samples) with global fallback.

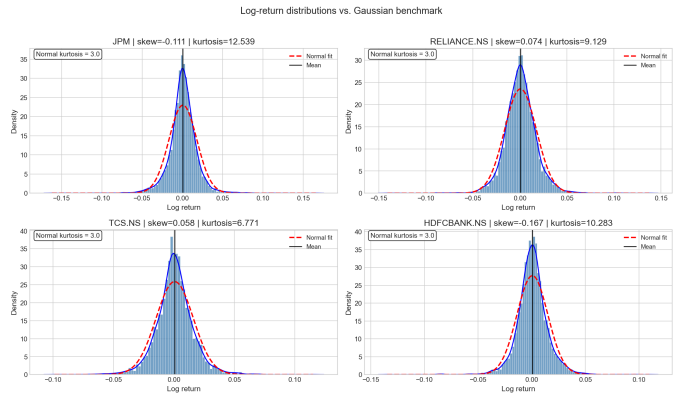

Fig. 1. Return distribution diagnostics (heavy tails and asymmetry) used to justify regime-aware modeling.

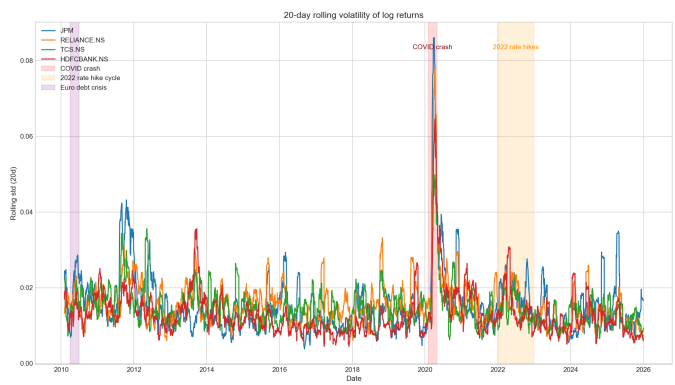

Fig. 2. 20-day rolling volatility (actual EDA output), motivating explicit HMM state labeling.

IV. Preliminary Results (XGBoost Baseline)

Table I reports ticker-level baseline metrics from the confirmed regime-stratified run.

Regime-stratified XGBoost achieves average DA of 52.08%. TCS shows the strongest risk-adjusted result (Sharpe 0.85). HDFCBANK and EPIGRAL remain weak on Sharpe, consistent with noisy regime transitions and higher idiosyncratic effects in smaller names.

EPIGRAL.NS was included as a small-cap generalization test. Although DA is 51.32%, the negative Sharpe (-0.56) indicates weak risk-adjusted behavior. This suggests OHLCV-only features are insufficient for small-cap settings and motivates multimodal signals (e.g., news sentiment and search-trend proxies) in Phase-2.

Table II summarizes the top-5 XGBoost feature-importance ranks for each ticker.

HMM_Regime appears in top-5 for HDFCBANK and EPIGRAL, providing direct evidence that regime labels carry predictive signal. Rolling_Corr_Index appears in top-5 for JPM and EPIGRAL, supporting the cross-asset context design.

TABLE III
Phase-1 Rubric to Evidence Mapping

Rubric Category	Evidence in This Submission
Literature Review	Sec. II; ARIMA/GARCH limitations, SOTA comparison, explicit gap synthesis [6], [7].
Dataset Quality & EDA	eda/eda.ipynb, figures in Fig. 1, Fig. 2.
Feature Engineering	features/feature_engineering.py; 36-column table, HMM regimes, cross-asset correlation.
Theoretical Rigor	HMM transition and emission equations in Sec. III-B.
Model Application	models/baseline_xgboost.py; walk-forward setup and Sec. IV results.
Code Quality / Reproducibility	Pipeline scripts in data/features/models, plus saved outputs in results.
Report Quality	IEEE format, equations, real figures, audited tables, and explicit phase scope.

TABLE I
Regime-Stratified XGBoost: Main Results

Ticker	RMSE	MAE	DA (%)	Sharpe
JPM	0.0194	0.0127	52.13	0.53
RELIANCE.NS	0.0180	0.0121	52.25	0.54
TCS.NS	0.0151	0.0106	53.17	0.85
HDFCBANK.NS	0.0165	0.0111	51.52	0.04
EPIGRAL.NS	0.0243	0.0166	51.32	-0.56
Average	0.0187	0.0126	52.08	0.28

TABLE II
Top-5 Features per Ticker (XGBoost Importance)

Ticker	R1	R2	R3	R4	R5
JPM	Momentum_60	BB_Lower	BB_Upper	Rolling_Corr_Index	Momentum_20
RELIANCE.NS	Garman_Klass_Vol	MACD	Return_Lag_2	BB_Position	ROC_10
TCS.NS	Momentum_20	BB_Upper	MACD_Hist	ROC_10	Momentum_60
HDFCBANK.NS	Momentum_60	Momentum_20	HMM_Regime	BB_Lower	MACD
EPIGRAL.NS	HMM_Regime	ROC_10	Momentum_20	MACD_Hist	Rolling_Corr_Index

V. Conclusion

Phase-1 is intentionally foundation-first: EDA, HMM regimes, engineered features, and a validated XGBoost baseline. This creates a strong technical base for later phases. The future architecture path (RAMT-style regime-gated multimodal transformer) remains a planned extension, not a claimed Phase-1 implementation.

Appendix: Phase 1 Rubric Mapping

References

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